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HUANENG POWER INTERNATIONAL: ACQUISITION OR THE PREDECESSOR METHOD? A DECISION ON A BUSINESS COMBINATION UNDER COMMON CONTROL

It was early 2024. Sara Lee, a freelance business analyst, received an interesting contract from one of her clients. She was asked to review how Huaneng Power International Inc. (HPI), a company incorporated in the People's Republic of China (PRC), had applied the acquisition method voluntarily in its International Financial Reporting Standard (IFRS) reporting. HPI's accounting choice was unique in that most internationally-listed Chinese companies used the predecessor method for IFRS reporting, consistent with their reporting under PRC GAAP (Generally Accepted Accounting Principles).

Lee had been asked to compare the impact of the acquisition and the predecessor methods on HPI and evaluate which of the two methods better reflected the economics of what is described as a 'business combination under common control' (BCUCC) under the IFRS. She also had to examine the strategic motivations for a BCUCC within the dynamic economic landscape and internationalisation of the power and utility industry in China. Lee was aware that the BCUCC was a common group restructuring arrangement, but was often insufficiently dealt with in accounting and business curricula. As an external investor of HPI, she was particularly interested in this deal and decided to start working on the review immediately.

The BCUCC

On January 1, 2015 (the acquisition date[1]), HPI, a company incorporated in the PRC completed the acquisition and obtained control of 10 companies from China Huaneng Group Co., Ltd (Huaneng Group) and another company controlled by Huaneng Group called the Huaneng International Power Development Corporation (HIPDC).

HPI was listed on the Hong Kong Stock Exchange (HKEX), and the Shanghai Stock Exchange (SSE), and its American Depositary Receipt (ADR) was listed on the New York Stock Exchange (NYSE). The transfer of the 10 entities from Huaneng Group and HIPDC to HPI was conducted according to the equity transfer agreement entered on October 13, 2014, by the three entities. The purchaser and sellers were all related parties engaged in similar principal activities.

The acquired entities were previously under the control of HIPDC (the direct controlling shareholder of HPI) and Huaneng Group (the ultimate parent company of HIPDC and HPI). As the combining entities were controlled by the same party (Huaneng Group) before and after the combination and that control was not transitory, the acquisition constituted a BCUCC as defined

1 The acquisition date of January 1, 2015, was disclosed in Note 5 of HPI "A" share 2015 annual report.

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by IFRS 3, *Business Combinations*[2], issued by the International Accounting Standards Board (IASB).

In a BCUCC event, the control of subsidiaries did not change from the perspective of the ultimate shareholders. In HPI's case, Huaneng Group continued to remain as the ultimate parent, and HIPDC as the intermediate parent after the transfer of their subsidiaries to HPI (refer to **Exhibit 1** for the relationship among HPI, Huaneng Group, HIPDC and the 10 acquired entities before and after the BCUCC event).

Both the transferring and the receiving entities and the acquired entities in the BCUCC were involved in similar principal activities (refer to Tables 1 and 2 in **Appendix A**). The similarity in principal activities presented Lee with an interesting question on the strategic motivations for transferring subsidiaries from the ultimate parent to another subsidiary.

As of January 2024, while accounting for BCUCC was not governed by IFRS 3 or any accounting standard, it was on the agenda of the IASB for the development of a future standard.[3] Being 'scoped out' of IFRS 3, reporting entities in the IFRS reporting domain typically used either the acquisition method under IFRS 3 or the predecessor method to account for a BCUCC. As such, financial reporting practices were diverse and the impact of the reported information on economic decisions by user groups could be significant.

Lee found the HPI case interesting because the company presented two sets of financial statements that complied with different listing requirements. Specifically, one set was prepared under PRC GAAP for its listing on SSE, and the other was prepared using IFRS for its listing on HKEX and NYSE. For PRC GAAP reporting, HPI used the predecessor method to account for the BCUCC according to PRC GAAP requirements. For IFRS reporting, it had voluntarily chosen the acquisition method from IFRS 3. This unique setting allowed her to compare and evaluate the two accounting methods for the same acquisition (refer to **Exhibit 2** for more details on accounting guidelines).

Before she could start her comparative analysis, Lee researched more on the background of HPI and its parent company, Huaneng Group.

Business Overview of HPI

HPI was one of Asia's largest listed power producers, with annual domestic power production ranging from 302,300 GWh (gigawatt hours) to 430,500 GWh. It mainly invested in, constructed, operated, and managed power plants. The most important segment of HPI was the Sale of Power and Heat, which made up more than 95% of its revenues (based on financial information disclosed in HPI's H-shares annual reports from 2015 to 2020). The remaining segments of HPI were the Sale of Material, Lease Income, Port Service, Transport Services, and Others. These segments grew from 0.6% to 4.7% of total revenues from 2016 to 2020.

2 IFRS, "IFRS 3 Business Combinations", https://www.ifrs.org/issued-standards/list-of-standards/ifrs-3-business-combinations/, accessed February 2024.
3 IFRS, "Business Combinations under Common Control", https://www.ifrs.org/projects/work-plan/business-combinations-under-common-control/, accessed February 2024.

HPI mainly operated in the conventional energy sector. It generated power from a variety of fuels including coal, gas, wind, water (hydro) and solar, but most of its power generation was coal-based. In 2015, 94.26% of HPI's power generated was coal-fired (302,126 GWh) and 4.61% was from a combined cycle using gas (14,787 GWh), while wind power, hydropower and photovoltaic (solar) together only accounted for 1.12%.[4] Its power generation operations were located across China[5] and its overseas market in Singapore.

Net profit margin of HPI ranged from 1.2% to 9.9% during the 2011-2020 period, with an average net profit margin of around 6%. In terms of sales, HPI experienced some fluctuations over the years and the 2014-2016 period was challenging for the company. Revenues dipped by 6.3% (year-on-year) in 2014 and 11.71% in 2016. There appeared to be a modest revenue growth of 2.79% in 2015. However, HPI would have reported another revenue dip in 2015 without the revenue contributed by 10 entities acquired through the BCUCC. Revenue growth was most significant from 2017 to 2019, particularly in 2017 (refer to **Exhibit 3**).[6]

Growth Strategy of HPI

As China developed the power market, total energy production in China continued to expand over the years, increasing 54.85% from 4,730,600 GWh in 2011 to 7,325,300 GWh in 2020.[7] HPI was active in maintaining its leading position and market share by expanding operations and enhancing efficiency.[8]

HPI stated in its 2015 20-F filing[9] that an important component of HPI's growth strategy was to acquire existing power plants from Huaneng Group, HIPDC and other companies on commercially reasonable terms. HPI also stated that the timing of such acquisitions depended on its ability to obtain financing and PRC governmental approval of the financing terms.

HPI had consistently grown its total assets every year during the 2011-2020 period, supporting its strategy to grow assets and maintain market share. Assets grew most significantly in 2015 at 12.24% (consistent with the operational expansion through the BCUCC) and in 2017 by 25.97% (refer to **Exhibit 4**).[10]

HPI stated that it strove to improve the synergy within the operations of its group of companies and to expand the market for its sales by consolidating its leading position in the conventional energy sector.[11] The BCUCC from HIPDC and Huaneng Group helped to achieve growth and efficiency objectives more expeditiously than was possible under organic means.

4 Data obtained from H share 2015 annual report.
5 HPI's domestic operations coverage include the Northeast China Grid, the Northern China Grid, the Northwest China Grid, the Eastern China Grid, the Central China Grid, the Southern China Grid (HPI ADR 20F 2015, Page 63).
6 Financial data was obtained from annual reports of H share of HPI (HKEX 902, reported under IFRS).
7 China energy market data was based on statistical publications by the China Electricity Council (english.cec.org.cn and chinaenergyportal.org). The authors cross-checked the data for verification.
8 HPI ADR 20-F 2015, Page 90. To note, 20-F filing refers to: "SEC Form 20-F is an annual report filing for non-U.S. and non-Canadian companies that have securities trading in the U.S. SEC Form 20-F helps standardise the reporting requirements of foreign-based companies. The company must also make the report available to shareholders through the company's website". Source: https://www.investopedia.com/terms/s/sec-form-20-f.asp.
9 HPI ADR 20-F 2015, Page 5.
10 HPI financial data were obtained from HPI H share annual reports.
11 HPI H share 2015 annual report, Page 32.

Amid the rapid development of the electricity market in China, HPI's growth strategy allowed the firm to maintain its market share, which remained relatively stable at around 5-6% (refer to **Exhibit 5**). The market share increased slightly in 2015, mainly due to the BCUCC.

Business Overview of Huaneng Group[12]

Huaneng Group, HPI's ultimate parent, was one of China's largest state-owned electricity generation companies, established with the approval of the country's State Council. Huaneng Group was the first among Chinese power generation enterprises to join the ranks of the Fortune Global 500 and was ranked 266 in 2020. Huaneng Group had 130,000 employees and owned 51 secondary units, more than 460 tertiary enterprises, and five public companies.[13] At the end of 2020, the total installed capacity of its wholly-owned and majority-owned power plants was 197.33 GWh, and the production capacity of coal was 85 million tons per year. Huaneng Group's total assets were close to 1.2 trillion RMB[14].

HPI was an important subsidiary of the Huaneng Group. First, it was Huaneng Group's only subsidiary with an international listing. Second, HPI accounted for more than one-third of Huaneng Group's assets (36.52% in 2020).[15] Third, it contributed to around half of Huaneng Group's core operation of power generation (refer to **Exhibit 6**).[16]

Intensifying Competition in China's Power Market

To create a more open and fair power market, the PRC Government started to reform the power industry in 2002. Five state-owned power generation companies, including Huaneng Group, were created following the break up of the former State Power Corporation of China (refer to **Exhibit 7** for details on the five major companies and their major listed subsidiaries).[17] Except for China Guodian Corporation, all the other companies had internationally-listed subsidiaries. It was noteworthy that in 2017, China Guodian was acquired by Shenhua Group to form a new entity named China Energy Investment Corporation. This new entity replaced China Guodian as one of the five major firms in the power industry.[18] One caveat of comparison after 2017 was that China Energy had significant operations not just in power generation but also in coal supply.

Competition was strong. In its 20-F filing of 2015, HPI stated that Huaneng Group had been facing competition from the other four state-owned companies (the non-listed parents).[19]

12 Company Website, https://www.chng.com.cn/en/companyoverview.html, China Huaneng, accessed February 2024.
13 Other than HPI, the other four public subsidiaries were listed on Shanghai Stock Exchange and Shenzhen Stock Exchange. They were Inner Mongolia Mengdian Huaneng Thermal Power Corp. Ltd., Shandong Xinneng Taishan Power Generation Co., Ltd., Huaneng Lancang River Hydropower Inc., and China Great Wall Securities Co., Ltd.
14 US$1 = RMB 6.9 (average for 2020), https://www.exchangerates.org.uk/USD-CNY-spot-exchange-rates-history-2020.html, accessed February 2023.
15 Financial data of Huaneng Group was reported under PRC GAAP only. When comparing HPI with Huaneng Group, the authors also used HPI's data prepared under PRC GAAP to match with the accounting of its parent company.
16 Data for Huaneng Group was obtained from the Prospectus when it issued debt: China Huaneng Group Prospectus 2021528, Page 46, China Huaneng Group Prospectus 2018613, Page 33, China Huaneng Group Prospectus 2015317, Page 36. Data for HPI was obtained from HPI H share AR 2011-2019.
17 HPI ADR 20-F 2015, Page 4.
18 David Reid, "China Just Approved the World's Largest Power Company with more than 225 Gigawatts of Capacity", CNBC, August 28, 2017, https://www.cnbc.com/2017/08/28/china-approves-merger-of-guodian-and-shenhua-to-create-worlds-largest-power-company.html, accessed February 2024.
19 HPI ADR 20-F 2015, Page 59.

Meanwhile, HPI had been competing with the listed subsidiaries of the other four state-owned companies.

Lee compared key financial metrics of HPI and its competitors listed on HKEX which produced IFRS financial statements (refer to **Exhibit 8** for asset and revenue comparison of HPI and its competitors).[20] GD Power Development Company was not included as it only reported under PRC GAAP. Datang International exceeded HPI in terms of total assets from 2012 to 2014, the period before the BCUCC. However, HPI was able to grow its assets more rapidly than all competitors in 2015 and subsequent periods. HPI consistently generated the highest revenues, significantly above its competitors.

Lee also compared the non-listed parents which produced annual reports under PRC GAAP (refer to **Exhibit 9** for an asset and revenue comparison of non-listed parents). It showed that Huaneng Group was the largest company in terms of both assets and revenues, until it was overtaken by China Energy Investment Corp in 2017.

The Morning After: Benefits Following the BCUCC

Multifaceted Growth

The BCUCC in 2015 led to significant business expansion for HPI across all fronts, from assets and production scale to revenues, profits, growth potential, and geographical coverage.

Assets and production: Total assets increased by RMB 28,758.6 million on the acquisition date, representing an increase of 10.45% of HPI's total assets before the BCUCC transaction (RMB 275,171.7 million), while net assets increased by 8.31%. The assets acquired through the BCUCC were directly linked to the increased operational scale. The power generated by HPI in 2015 was 320,529 GWh, while 42,742 GWh came from the acquired power plants, representing 13.3% of the total (refer to **Exhibit 10**). The greater generation capacity helped HPI maintain or grow its market share. Moreover, greater synergies and economies of scale were expected from the consolidation of assets and operating expertise.

Profits and revenues: For the year ended December 31, 2015, the acquired business contributed 14.95% to HPI's consolidated profits (RMB 2,581 million out of RMB 17,259 million) and 12.01% to HPI's consolidated revenue (RMB 15,477 million out of RMB 128,905 million).[21] Without the revenues contributed by the acquired companies, HPI would have reported a decrease in revenue in 2015.

Growth potential: At the time of BCUCC, three acquirees[22] had power generation capacity under construction. These capacities under construction (2,300 MW) represented 3.1% of the future additional power generation capacity for HPI.

20 Except for HPI, the authors obtained all the financial data of listed subsidiaries and six non-listed parents from Bloomberg. HPI financial data was obtained from HPI H share annual reports.
21 HPI H share 2015 annual report, Note 40.
22 The three acquirees were Anyuan Power, Jingmen Thermal Power and Yingcheng Thermal Power.

Geographical coverage: BCUCC allowed HPI to expand its footprint (refer to **Exhibit 11** for a map showing the existing and acquired plants). More importantly, it enabled HPI to gain a foothold in the power generation markets of Hainan and Anhui.[23]

Reduced Intragroup Competition

Guotai Junan Capital Limited (the financial adviser to HPI's independent board and independent shareholders for this BCUCC transaction) asserted that the BCUCC in 2015 fulfilled Huaneng Group's goal to eliminate intragroup competition and support the development of HPI.[24] Before the BCUCC, there was competition between HPI and its shareholders (HIPDC and Huaneng Group) because they operated in similar businesses and had power plants located in the same regions. In 2010, HPI received an undertaking from Huaneng Group to avoid business competition within the group.[25] In the undertaking, Huaneng Group pledged that HPI would be the only platform to integrate the conventional energy business in the group. Furthermore, Huaneng Group pledged that some conventional energy business of the group would be injected into HPI upon meeting certain conditions. In its 2015 annual report, HPI stated that the BCUCC transaction further reduced business competition between HPI and its controlling shareholder.[26]

Consolidation of Operations and Unlocking Value

The anticipated easing of demand and supply in the electricity market also provided incentives to consolidate operations. China Electricity Council predicted that total electricity consumption nationwide in 2016 would only grow by 1% to 2%. The electricity market supply would be sufficient to meet the demand nationwide, with several regions (Northeast and Northwest) and several provinces having abundant surplus capacity.[27] After the BCUCC, operations from more locations were integrated. This enabled HPI to conduct more coordinated decisions (for example, achieving economies of scale for each power plant) and to operate at a more efficient level.

Huaneng Group and HIPDC on the other hand could unlock the value created in the 10 companies (the difference between fair value and carrying amounts of identifiable net assets) through the BCUCC. The BCUCC also effectively allowed them to realise goodwill (previously internally generated goodwill) of an amount that was close to RMB 1.17 billion in cash. Most of the goodwill realised could be attributed to Hainan Power and Wuhan Power.

Additionally, the BCUCC enabled Huaneng Group to play a more strategic role in the management of subsidiaries. The transfer of control over power plants, which Huaneng Group previously had to manage, freed up the capacity for the Group to conduct higher-level management and strategic decision-making. After the BCUCC, HPI entered into a Framework Agreement and Entrusted Management Agreement with Huaneng Group. Under these agreements, Huaneng Group would centralise functions and decisions relating to leasing, provision of coal and transport services, management of budgeting, finance, human relations, administration, legal, and internal control.[28]

23 HPI H share 2015 annual report, Page 18.
24 HPI's report to HKEX on November 13, 2014, DISCLOSEABLE AND CONNECTED TRANSACTIONS, Page 72
25 HPI H share 2015 annual report, Page 77.
26 Ibid.
27 HPI H share 2015 annual report, Page 32.
28 HPI ADR 20-F 2015, Page 104.

A Tale of Two Treatments: Deconstructing the BCUCC

Having studied the background of the BCUCC, Lee now began to analyse HPI's dual accounting treatment for its reporting. Because of different listing requirements, HPI had adopted the predecessor method under PRC GAAP for its domestic reporting and the acquisition method for the same BCUCC transaction under IFRS for its international reporting (refer to **Exhibit 12** for details on HPI's accounting policies on BCUCC for reporting of its different types of shares).

HPI's dual accounting treatments for BCUCC were not merely explained by the fact that HPI had both A-shares and H-shares. Even among Chinese companies with both A-shares and H-shares, it was extremely rare for a company to apply the acquisition method in its IFRS reporting. Most companies chose to apply the predecessor method in their H-shares' financial reports to have a consistent accounting policy with their controlling parties in the PRC (HKICPA, 2019).[29] Furthermore, the predecessor method had a more favourable impact on the consolidated financial statements and was easier to prepare than the acquisition method. Out of the 113 BCUCC transactions on HKEX during the 2013-2016 period, Lee found that only one company applied both the predecessor method and the acquisition method for the same BUCCC transaction.[30] That company was HPI.

The Acquisition Method under IFRS

In its consolidated financial statements prepared under IFRS reporting, HPI and its subsidiaries applied the acquisition method to the BCUCC in 2015. The total consideration was RMB 9,647,248,000 which was fully settled in cash in 2015 (refer to **Exhibit 13** for a summary of the key amounts presented by Note 40 of HPI's H-share 2015 financial statements applying the acquisition method under IFRS for the BCUCC).

Goodwill arising from the BCUCC was attributable to the synergies expected to arise after the acquisition, which mainly included goodwill arising from the acquisition of Hainan Power and Wuhan Power. In addition, HPI recognised the identifiable net assets of the 10 acquired subsidiaries at fair value to reflect the effective control and acquisition of the identifiable net assets. Although the transferring entities were related parties of HPI, the transaction was seen as an acquisition of control by HPI. Hence, as in a typical acquisition, identifiable net assets acquired would be recognised at fair value to reflect the purchase transaction and the gain of control over these assets. As the identifiable net assets were recognised at fair value on the acquisition date, HPI also recognised higher depreciation, amortisation and expense arising from the fair value uplift of these assets in subsequent post-acquisition periods. Under the acquisition method, comparative figures for HPI's consolidated financial statements were not restated to include consolidation of the 10 entities as control of these entities had not existed in past reporting periods.

Table 1 in **Appendix B** presents the carrying amounts as well as the fair values of identifiable net assets for each of the 10 entities, while Table 2 presents the consolidated statement of financial position as of December 31, 2014, immediately before the BCUCC.

29 Hong Kong Institute of Certified Public Accountants (HKICPA), "Common Control Combinations – How they were Reported in Hong Kong", 2019, https://www.hkicpa.org.hk/en/Standards-and-regulation/Standards/Major-projects/Business-Combinations-under-Common-Control, accessed February 2024.
30 Ibid.

The Predecessor Method under PRC GAAP

According to the extant PRC GAAP[31], HPI measured the net assets of subsidiaries acquired at their carrying amounts (book values) and not fair values. For simplicity, Lee assumed that the book values were those carrying amounts as reported in the legal entity financial statements of each subsidiary at the date of the BCUCC.

Under the predecessor method, HPI did not measure the identifiable net assets acquired at fair value, unlike the acquisition method under IFRS reporting. HPI also did not recognise any goodwill arising from the BCUCC. Instead, the difference between the carrying amounts of the net assets acquired and the consideration paid was adjusted to the equity accounts (typically capital reserves) of the acquirer (if the difference was a debit and exceeded capital reserves, the excess was adjusted against retained earnings). Furthermore, HPI retrospectively restated the reported comparative figures as if the acquired entities had been under the control of HPI since the beginning of the earliest year presented. Lastly, HPI's depreciation and amortisation after the acquisition were not adjusted upwards for any fair value uplift because identifiable net assets were carried at their carrying amounts (or book values) and not at fair value on the date of the BCUCC (refer to **Exhibit 14** for items reported under the predecessor method).

The predecessor method was less complex, resulting in lower processing costs. The numbers reported under the predecessor method were less subjective than fair values reported under the acquisition method, with a lower risk of errors and consequential litigation risks.

Post-Acquisition Changes after the BCUCC

As a result of recognising identifiable net assets at fair value under the acquisition method, HPI recognised a higher expense as and when the net assets were used or consumed (**Appendix B,** Table 3 shows the consolidated income statement prepared for the year ended December 31, 2015, prepared under IFRS reporting). However, under the predecessor method, HPI did not have to show a higher expense in post-acquisition periods arising from the difference between the fair value and the book value of an asset (typically referred to as the 'fair value uplift') as the assets were carried at book value. There was no need to follow through with the subsequent expensing or amortisation of the fair value differentials resulting in more positive income effects under the predecessor method than the acquisition method.

Since the receiving entity did not recognise goodwill under the predecessor method, the entity did not have to carry annual reviews for the impairment of goodwill with a positive impact on income. Generally, higher and less volatile income was thus reported under the predecessor method vis-à-vis the acquisition method with consequential advantages for the receiving entity and its management. However, these advantages did not necessarily benefit the external investors of the receiving entity who needed information on the impact of the acquisition on synergies and future cash flows of the receiving entity.

31 More complex accounting for BCUCC in PRC GAAP was in force before January 1, 2007.

Conclusion

Lee realised that in many financial reporting situations, a reporting entity needed to manage the tension between maximising the benefits of information to the investors and minimising costs to the reporting entity. Despite the less favourable and more costly impact of the acquisition method on the reporting entity, HPI had voluntarily adopted the acquisition method to report a BCUCC transaction for its IFRS financial statements. She mused: Was that a good accounting choice? What were the possible strategic motivations for the BCUCC?

Which of the two methods – the acquisition or the predecessor method – provided more relevant information to the external investors of HPI, an internationally-listed company, on the effects of the BCUCC? She believed her conclusions would help the IASB decide whether to permit the receiving entity to use either the acquisition or the predecessor method.

APPENDIX A

Table A1: Background Information of Transferring and Receiving Entities in the BCUCC

Company	Role in transfer	Relationship to HPI	Principal activities
HPI	Purchaser		Huaneng Power International Inc mainly develops, constructs, operates and manages large-scale power plants in China. It is one of the largest listed power producers in China.
Huaneng Group	Seller	Ultimate controlling shareholder and ultimate parent of HPI	Huaneng Group is principally engaged in the operation and management of industrial investments; the development, investment, construction, operation, and management of power sources; the organisation of the generation and sale of (thermal) power; and the development, investment, construction, production, and sale of products in relation to energy, transportation, new energy, and environmental protection industries.
HIPDC	Seller	Direct controlling shareholder or intermediate parent[32] of HPI	HIPDC is a Sino-foreign equity joint venture approved by the State Council to develop, construct, and operate power plants across China.

Source: HPI's report to HKEX on November 13, 2014, DISCLOSEABLE AND CONNECTED TRANSACTIONS, Page 7.

32 HIPDC was referred as the "direct controlling shareholder" in HPI's report to HKEX on November 13, 2014, DISCLOSEABLE AND CONNECTED TRANSACTIONS, Page 7.

Table A2: Background Information of Transferred Entities in the BCUCC

	Acquired business	**Acquired from**	**Equity interests acquired**	**Business nature and scope of operation**
1.	Hainan Power	Huaneng Group	91.8%	Construct power plants, new energy development, power plant engineering and equipment maintenance service
2.	Wuhan Power	Huaneng Group	75%	Power generation; comprehensive utilisation of fly ash
3.	Suzhou Thermal Power	Huaneng Group	53.45%	Power generation, steam production; mechanical and electoral equipment, pipeline installation, maintenance service; hot water, fly ash sales
4.	Dalongtan Hydropower	Huaneng Group	97%	Hydropower development, production and management of electric power urban water supply
5.	Hualiangting Hydropower	Huaneng Group	100%	Generation and transfer of power supply, water supply and irrigation
6.	Chaohu Power	HIPDC	60%	Construction and operation of power plants and related construction projects
7.	Ruijin Power Generation	HIPDC	100%	Construction and operation of power plants and related construction projects
8.	Anyuan Power	HIPDC	100%	Construction and operation of power plants and related construction projects
9.	Jingmen Thermal Power	HIPDC	100%	General business projects relating to thermal power, power development, power services
10.	Yingcheng Thermal Power	HIPDC	100%	Construction and operation of power plants and related construction projects

Source: HPI H share 2015 annual report, Note 9 and Note 40.

APPENDIX B

Table B1: Information on Identifiable Net Assets of Acquired Companies Through the BCUCC on the Acquisition Date
(Amounts Expressed in Thousands of RMB)

Identifiable assets (liabilities)	Hainan Power Fair value	Hainan Power Acquiree's carrying amount	Wuhan Power Fair value	Wuhan Power Acquiree's carrying amount	Suzhou Thermal Power Fair value	Suzhou Thermal Power Acquiree's carrying amount	Dalongtan Hydropower Fair value	Dalongtan Hydropower Acquiree's carrying amount
Property, plant and equipment	8,198,754	7,347,394	4,723,925	3,929,268	626,551	552,971	303,255	303,255
Land use rights	1,328,452	412,525	1,312,397	186,236	81,075	20,104	12,420	12,420
Deferred income tax assets	0	109,888	0	3,727	0	2,746	0	0
Investment in an associate	136,244	136,244	0	0	0	0	0	0
Available-for-sale financial assets	0	0	0	0	33	33	0	0
Other non-current assets	26,823	25,180	0	0	373	373	0	0
Inventories	167,476	167,476	234,750	234,750	25,508	25,508	92	92
Receivables and other current assets	677,902	677,902	459,640	459,640	37,336	37,336	3,451	3,451
Payables and other current liabilities	(586,322)	(586,322)	(480,956)	(480,956)	(95,216)	(95,216)	(9,959)	(9,959)
Salary and welfare payables	(74,801)	(74,801)	(20,336)	(20,336)	(4,270)	(4,270)	-	-
Borrowings	(5,380,813)	(5,380,813)	(2,844,530)	(2,844,530)	(230,000)	(230,000)	(199,187)	(199,187)
Other non-current liabilities	(2,700)	(2,700)	(40,037)	(40,037)	(7,176)	(7,176)	(760)	(760)
Deferred income tax liabilities	(332,344)	0	(476,477)	0	(30,892)	0	0	0
Bank balances and cash	269,420	269,420	241,796	241,796	25,718	25,718	3,709	3,709
Total identifiable net assets/(liabilities)	4,428,091	3,101,393	3,110,172	1,669,558	429,040	328,127	113,021	113,021

Identifiable assets (liabilities)	Hualiangting Hydropower Fair value	Hualiangting Hydropower Acquiree's carrying amount	Chaohu Power Fair value	Chaohu Power Acquiree's carrying amount	Rujin Power Fair value	Rujin Power Acquiree's carrying amount	Anyuan Power Fair value	Anyuan Power Acquiree's carrying amount
Property, plant and equipment	4,224	4,224	3,170,347	2,598,201	2,028,220	1,942,631	3,330,391	3,323,710
Land use rights	32,707	32,707	4,603	3,344	52,269	53,909	89,264	89,264
Deferred income tax assets	0	0	0	0	0	0	0	0
Investment in an associate	0	0	0	0	0	0	0	0
Available-for-sale financial assets	105	105	0	0	0	0	0	0
Other non-current assets	207	207	230	230	294	294	83,030	83,030
Inventories	359	359	131,697	131,697	115,013	115,013	1,984	1,984
Receivables and other current assets	3,090	3,090	252,004	252,004	184,600	184,600	29,401	29,401
Payables and other current liabilities	(32,021)	(32,021)	(275,897)	(275,897)	(161,977)	(161,977)	(700,972)	(700,972)
Salary and welfare payables	(2,387)	(2,387)	(916)	(916)	(1,596)	(1,596)	(56,984)	(56,984)
Borrowings	(10,000)	(10,000)	(1,690,220)	(1,690,220)	(1,868,750)	(1,868,750)	(3,082,500)	(3,082,500)
Other non-current liabilities	-	-	(932)	(932)	(110)	(110)	(25,288)	(25,288)
Deferred income tax liabilities	0	0	(143,351)	0	(20,987)	0	(1,670)	0
Bank balances and cash	11,470	11,470	30,210	30,210	53,381	53,381	188,402	188,402
Total identifiable net assets/(liabilities)	7,754	7,754	1,477,775	1,047,721	380,357	317,395	-144,942	-149,953

Identifiable assets (liabilities)	Jingmen Thermal Power Fair value	Jingmen Thermal Power Acquiree's carrying amount	Yingcheng Thermal Power Fair value	Yingcheng Thermal Power Acquiree's carrying amount	Total Fair value	Total Acquiree's carrying amount
Property, plant and equipment	2,467,152	2,465,509	1,609,179	1,609,642	26,461,998	24,076,805
Land use rights	6,615	6,784	76,638	71,950	2,996,440	889,243
Deferred income tax assets	0	0	0	0	0	116,361
Investment in an associate	0	0	0	0	136,244	136,244
Available-for-sale financial assets	0	0	0	0	138	138
Other non-current assets	88	88	0	0	111,045	109,402
Inventories	30,529	30,529	0	0	707,408	707,408
Receivables and other current assets	102,822	102,822	1,105	1,105	1,751,351	1,751,351
Payables and other current liabilities	(379,138)	(379,138)	(346,586)	(346,586)	(3,069,044)	(3,069,044)
Salary and welfare payables	(868)	(868)	-	-	(162,158)	(162,158)
Borrowings	(1,886,834)	(1,886,834)	(1,162,000)	(1,162,000)	(18,354,834)	(18,354,834)
Other non-current liabilities	-	-	(54,977)	(54,977)	(131,980)	(131,980)
Deferred income tax liabilities	(369)	0	(1,057)	0	(1,007,147)	0
Bank balances and cash	61,508	61,508	87,072	87,072	972,686	972,686
Total identifiable net assets/(liabilities)	401,505	400,400	209,374	206,206	10,412,147	7,041,622

Source: Note 40, HPI H Share 2015 Annual Report

Table B2: Comparative Statement of Financial Position of HPI as of December 31, 2014

(Amounts Expressed in Thousands of RMB)

	31-Dec-14
ASSETS	
Non-current assets	
Property, plant and equipment	188,379,057
Investments in associates and joint ventures	17,626,910
Available-for-sale financial assets	4,333,377
Land use rights	4,953,844
Power generation license	3,720,959
Mining rights	1,922,655
Deferred income tax assets	884,274
Derivative financial assets	40,598
Goodwill	11,725,555
Other non-current assets	3,719,255
Total non-current assets	237,306,484
Current assets	
Inventories	6,702,274
Other receivables and assets	3,411,720
Accounts receivable	14,881,963
Derivative financial assets	261,135
Bank balances and cash	12,608,192
Total current assets	37,865,284
Total assets	275,171,768
EQUITY AND LIABILITIES	
Capital and reserves attributable to equity holders of the Company	
Share capital	14,420,383
Capital surplus	19,622,199
Surplus reserves	7,196,349
Currency translation differences	(1,194,819)
Retained earnings	30,085,379
	70,129,491
Non-controlling interests	14,653,215
Total equity	84,782,706
Non-current liabilities	
Long-term loans	57,638,458
Long-term bonds	22,725,535
Deferred income tax liabilities	1,810,755
Derivative financial liabilities	649,513
Other non-current liabilities	2,718,680
Total non-current liabilities	85,542,941
Current liabilities	
Accounts payable and other liabilities	27,035,864
Taxes payables	1,858,024
Dividends payable	431,681
Salary and welfare payables	171,262
Derivative financial liabilities	832,727
Short-term bonds	18,244,806
Short-term loans	43,529,004
Current portion of long-term loans	7,392,433
Current portion of long-term bonds	5,020,760
Current portion of other non-current liabilities	329,560
Total current liabilities	104,846,121
Total liabilities	190,389,062
Total equity and liabilities	275,171,768

Source: HPI H-share 2015 Annual Report

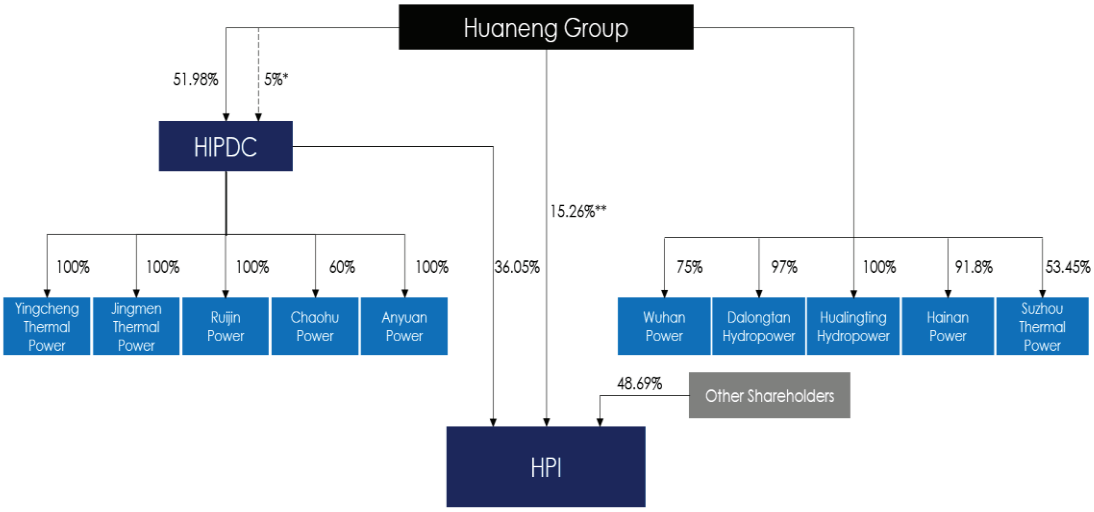
Table B3: Consolidated Income Statement Prepared Under IFRS for the Year Ended December 31, 2015
(Amounts Expressed in Thousands of RMB)

	Year ended December 31, 2015
Operating revenue	128,904,873
Tax and levies on operations	(1,157,760)
Operating expenses	
Fuel	(59,242,367)
Maintenance	(4,556,361)
Depreciation	(14,411,632)
Labour	(7,751,551)
Service fees on transmission and transformer facilities of HIPDC	(140,771)
Purchase of electricity	(3,581,517)
Others	(8,919,988)
Total operating expense	(98,604,187)
Profit from operations	29,142,926
Interest income	160,723
Financial expenses, net	
Interest expense	(7,945,734)
Exchange gain and bank charges, net	(24,336)
Total financial expenses, net	(7,970,070)
Share of profits less losses of associates and joint ventures	1,525,975
(Loss)/gain on fair values changes of financial assets/liabilities	(16,742)
Other investment income	115,238
Profit before income tax expense	22,958,050
Income tax expense	(5,698,943)
Net profit	17,259,107

Source: HPI H-share 2015 Annual Report

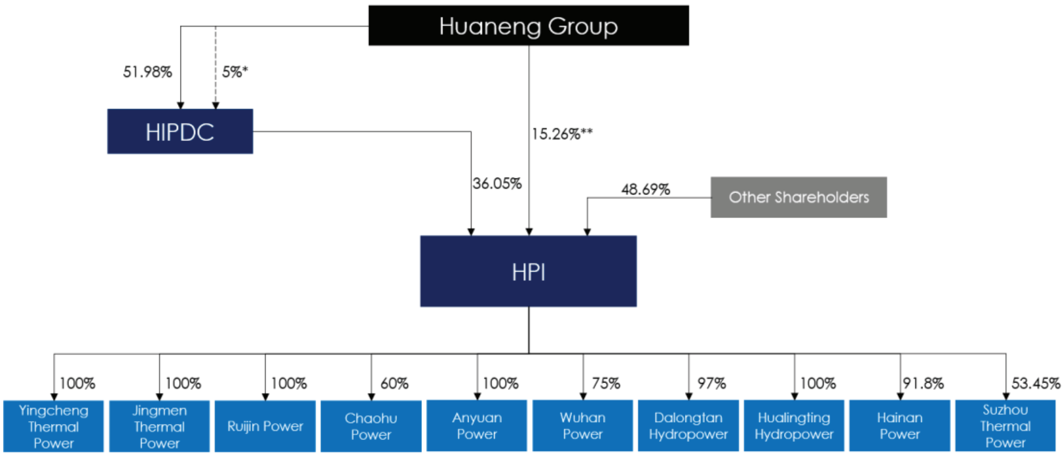
EXHIBIT 1: SHAREHOLDING STRUCTURE BEFORE AND AFTER THE BCUCC IN 2015

Before BCUCC



* Huaneng Group, through Huaneng HK, indirectly holds 100% of Pro-Power Investment Limited while Pro-Power Investment Limited holds a 5% interest in HIPDC. Therefore, Huaneng Group holds a 5% indirect interest in HIPDC.
** Huaneng Group holds an 11.06% direct interest in HPI. It also holds 3.37%, 0.04% and 0.79% indirect interests in HPI through Huaneng HK, Huaneng Capital Services Co. Ltd. and China Huaneng Finance Corporation Limited, respectively.

After BCUCC



Source: Constructed based on the disclosure in HPI's report to HKEX on November 13, 2014. As disclosed in that report, Huaneng Group's shareholding percentages in HIPDC and HIPDC's shareholding percentage in HPI reflect the relationship as of September 30, 2014. These percentages proxy for shareholding percentages on the acquisition date of January 1, 2015.

EXHIBIT 2: ACCOUNTING FOR A BCUCC

There were some critical issues with respect to accounting for a BCUCC. One key question was related to how the receiving entity (the entity which acquired control of the subsidiaries) should recognise and measure the business combination. In this case, the ultimate parent company, Huaneng Group, and HPI's direct controlling shareholder, HIPDC, transferred the ownership of 10 subsidiaries to HPI. Huaneng Group continued to hold a controlling interest in HPI and the transferred subsidiaries. HPI was thus the 'receiving entity' while Huaneng Group and HIPDC were the 'transferring entities'.

In answering the key question, a primary consideration was related to the information that should be reported to the shareholders of HPI, the receiving entity. Since HPI was listed on three major stock exchanges in three different reporting domains, the information that HPI reported on its acquisition was vital to its global investors. In the absence of an IFRS requirement on the accounting for a BCUCC, HPI had two accounting methods to choose from: the acquisition method or the predecessor method (the predecessor method was also referred to as 'pooling of interests' or the 'book value' method). Under the acquisition method, the receiving entity recognised goodwill and net assets of the transferred entities at fair value. However, the predecessor method recognised only the assets and liabilities of the transferred entities at book value as previously carried by the transferring entity; in turn, goodwill was not recognised.[33]

In the context of group reporting, the IFRS Conceptual Framework provided a guiding principle that consolidated financial statements must provide useful information for stakeholders in their assessment of the prospects for future cash flows of the parent.[34] The intermediate parent's own consolidated financial statements were designed to provide information about the assets, liabilities, equity, income, and expenses of the intermediate parent and its subsidiaries. HPI was a separate reporting entity from Huaneng Group and HIPDC, with its own distinct third-party stakeholders in the international markets. The receiving entity's perspective was thus de-linked from the ultimate parent's perspective. Although there was no change in control from the ultimate parent's perspective, there was a change in control from the receiving entity's perspective. HPI shareholders needed to know how control over the newly-acquired entities would affect the future cash flows of HPI.

A principle-based approach could be applied to account for the BCUCC to meet the Conceptual Framework characteristics of faithful representation. This accounting method could be determined based on the economics of the BCUCC, the strategic motivations and the impact on the cash flows of the receiving entity.

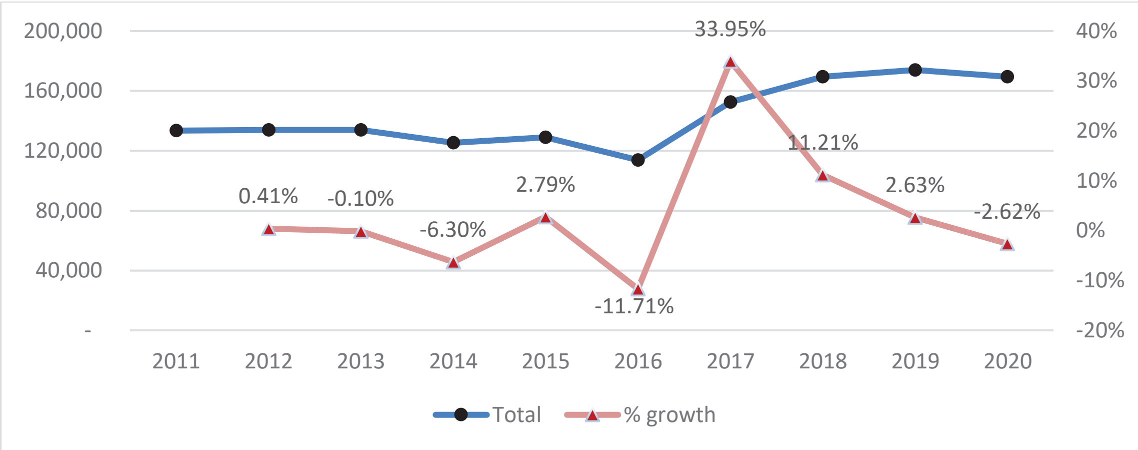
One approach was to evaluate the 'commercial substance' of the BCUCC.[35] If the BCUCC changed the amount, variability, and timing of the cash flows of the receiving entity, the recognition of goodwill and fair values of identifiable net assets may better reflect the expected cash flows from these assets than their book values. Recognising goodwill would also reflect the

33 For further details, refer to Pearl Tan, Lim Chu Yeong, and Tracey Zhang, "Business Combinations under Common Control (Part 1)", Institute of Singapore Chartered Accountants (ISCA) Journal, March 2021, https://isca.org.sg/docs/default-source/isca-journal-archive/2021-archive/isca-mar-2021.pdf, accessed February 2024.
34 IFRS Foundation, "Conceptual Framework for Financial Reporting", https://www.ifrs.org/issued-standards/list-of-standards/conceptual-framework/, accessed February 2024.
35 For further details, refer to Pearl Tan, Lim Chu Yeong, and Tracey Zhang, "Business Combinations under Common Control (Part 1)", Institute of Singapore Chartered Accountants (ISCA) Journal, March 2021, https://isca.org.sg/docs/default-source/isca-journal-archive/2021-archive/isca-mar-2021.pdf, accessed February 2024.

economic substance of synergies arising from the new group of companies. To help identify the 'commercial substance' of the BCUCC, Lee could evaluate the motivations for setting up the BCUCC. Strategic motivations of a BCUCC would provide clues as to the likelihood of changes in the amount, variability, and timing of cash flows of the receiving entity.

There could be different motivations for setting up a BCUCC arrangement. For example, a BCUCC could restructure a group to focus on certain products or markets. This could improve organisational efficiency and effectiveness, enhance business synergies, and eliminate internal competition. A BCUCC may also be carried out to reduce tax when loss-making entities were merged with profitable entities. Finally, a BCUCC could be carried out to spin off a subgroup for listing purposes. This was part of a divestment plan to create, unlock, and monetise value for the ultimate parent company. The strategic motivations for a BCUCC provide indications of whether control was transitory or not.[36]

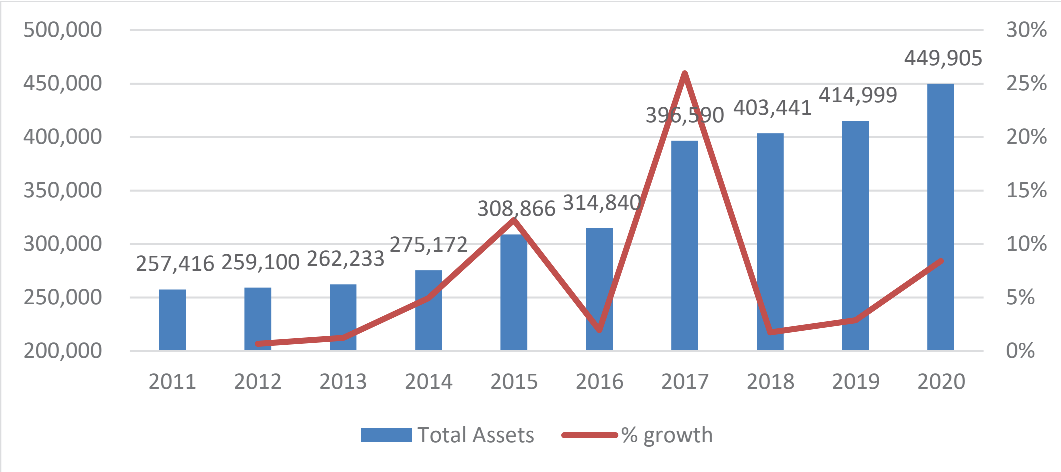
EXHIBIT 3: HPI: REVENUE AND REVENUE GROWTH (RMB MILLION)



Source: Constructed based on financial data obtained from Bloomberg (data are verified and corrected based on HPI's H-shares annual reports from 2011 to 2020), https://www.hpi.com.cn/sites/english/reportyh/publish.aspx, accessed February 2024.

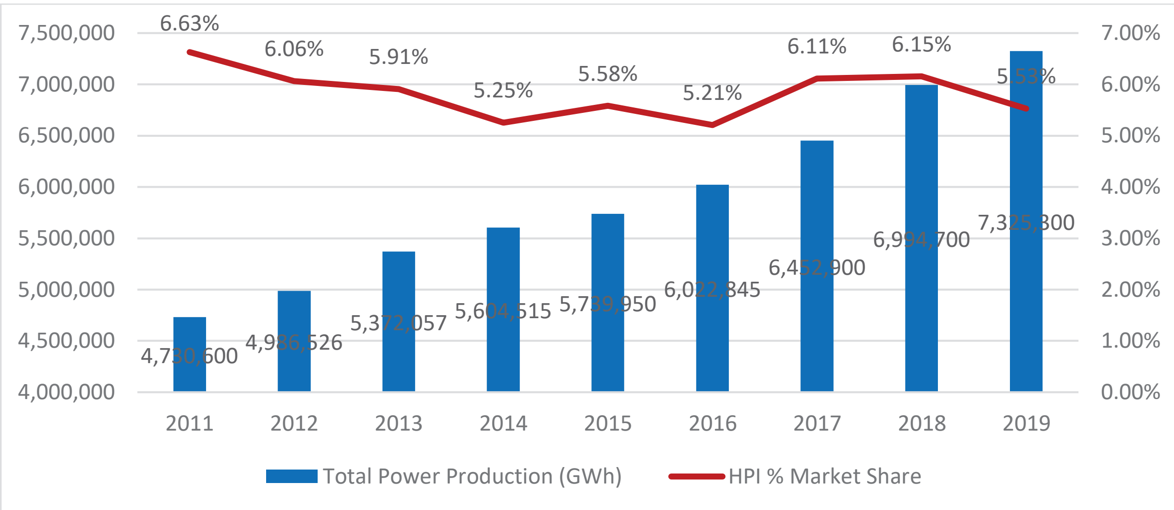
36 Ibid.

EXHIBIT 4: HPI: ASSETS AND ASSETS GROWTH (RMB MILLION)



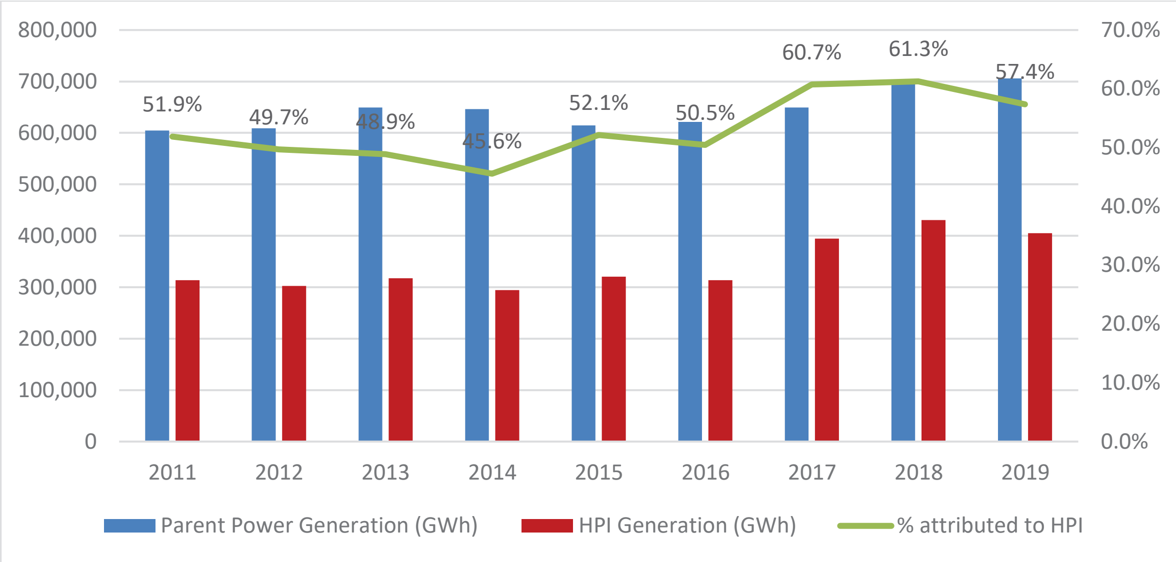
Source: Constructed based on financial data obtained from Bloomberg (data are verified and corrected based on HPI's H-shares annual reports from 2011 to 2020), https://www.hpi.com.cn/sites/english/reportyh/publish.aspx, accessed February 2024.

EXHIBIT 5: TOTAL ENERGY PRODUCTION IN CHINA AND HPI'S MARKET SHARE IN CHINA



Source: Data for total energy production in China is extracted from www.chinaenergyportal.org, and HPI's market share is constructed based on its energy production data from HPI's H-shares 2015 annual report from 2011 to 2019, https://www.hpi.com.cn/sites/english/reportyh/publish.aspx, accessed February 2024.

EXHIBIT 6: POWER GENERATION BY HUANENG GROUP AND HPI



Source: Data obtained from Prospectus as Company regularly issues debt, including China Huaneng Group Prospectus 2021528, Page 46; China Huaneng Group Prospectus 2018613, Page 33; and China Huaneng Group Prospectus 2015317, Page 36, accessed February 2024.

EXHIBIT 7: STATE-OWNED POWER GENERATION COMPANIES[37]

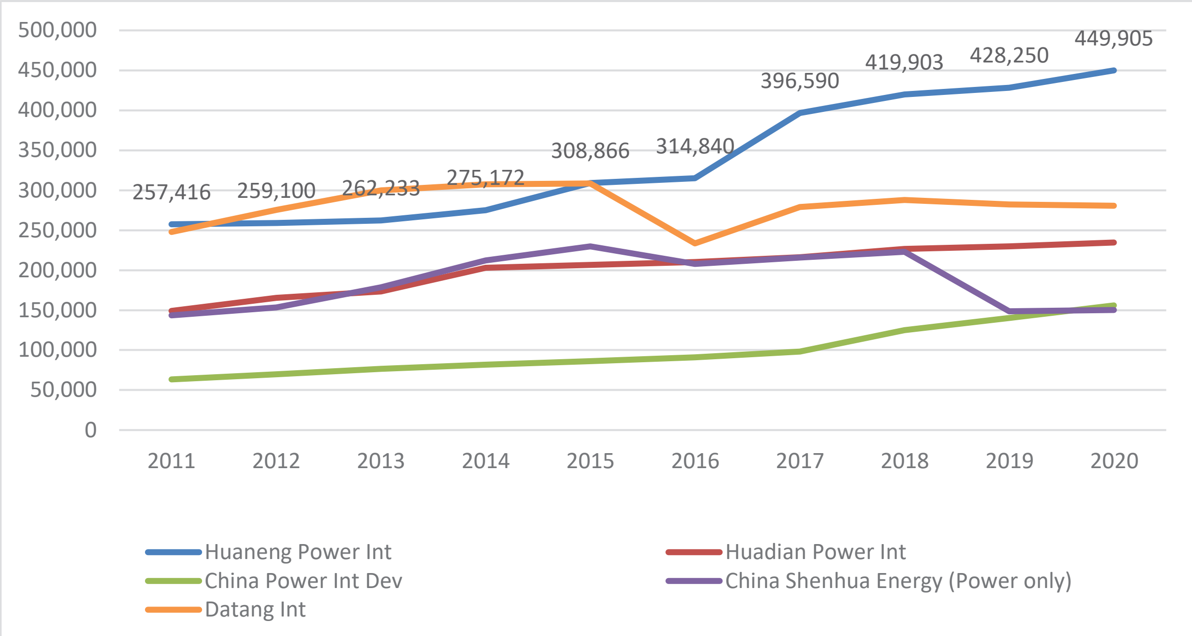
China Huaneng Group	• Parent of HPI (listed on HKEX and SSE)
China Datang Corporation	Parent of Datang International Power Generation Company (listed on HKEX and SSE)
China Guodian Corporation	Parent of GD Power Development Company (listed on SSE only)
China Huadian Group	Parent of Huadian Power International (listed on HKEX and SSE)
State Power Investment Corporation	Parent of China Power International Development (listed on HKEX only)
In 2017, China Guodian was replaced by	
China Energy Investment Corp	• Parent of GD Power Development Company (listed on SSE only) • Parent of China Shenhua Energy (listed on HKEX and SSE)

Source: Authors' compilation

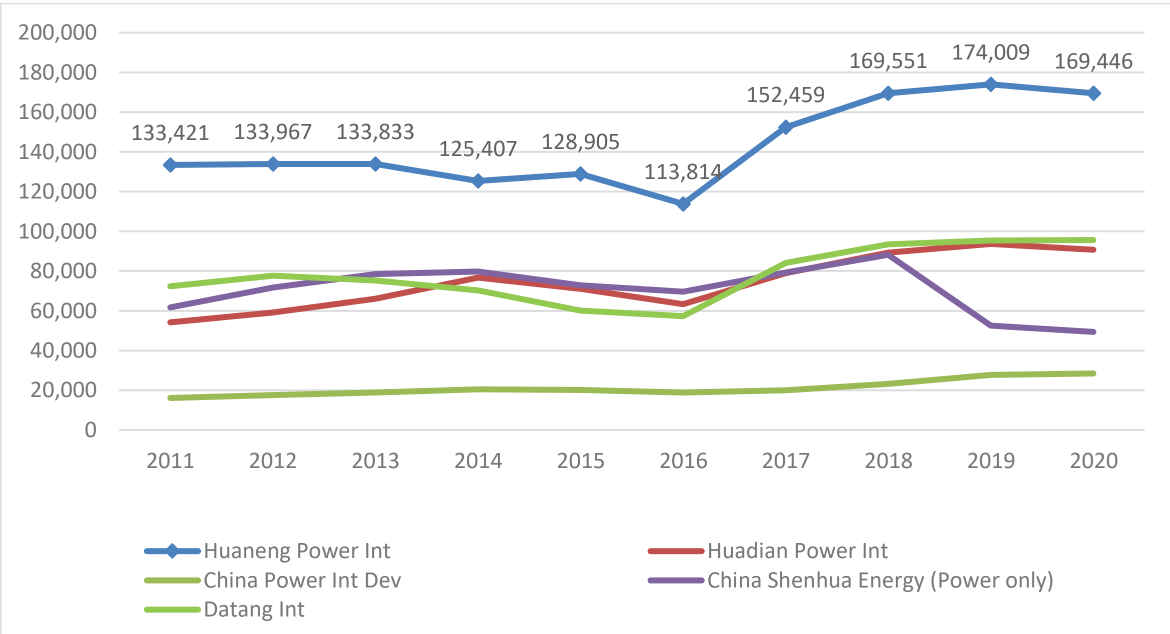
37 The following abbreviations were used in Exhibit 7. HKEX: Hong Kong Stock Exchange. SSE: Shanghai Stock Exchange.

EXHIBIT 8: ASSETS AND REVENUE COMPARISON OF HPI AND ITS COMPETITORS LISTED ON HKEX (IN RMB MILLION)

Assets Comparison of HPI and its Competitors Listed on HKEX (in RMB Million)



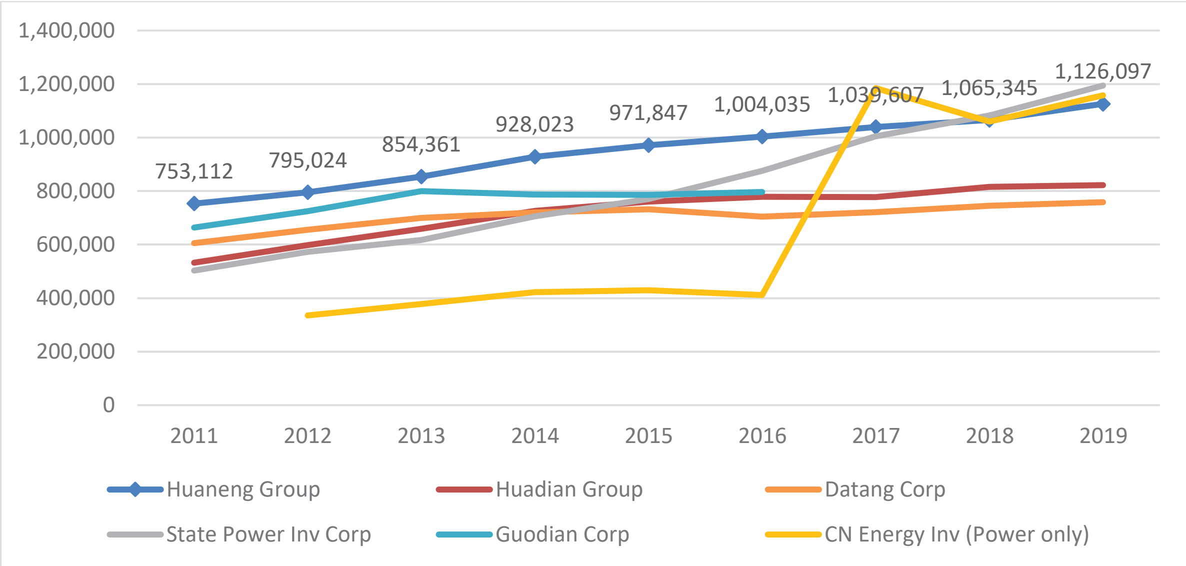
Revenue Comparison of HPI and its Competitors Listed on HKEX (in RMB Million)



Source: Constructed based on financial data obtained from Bloomberg Terminal, accessed February 2024.

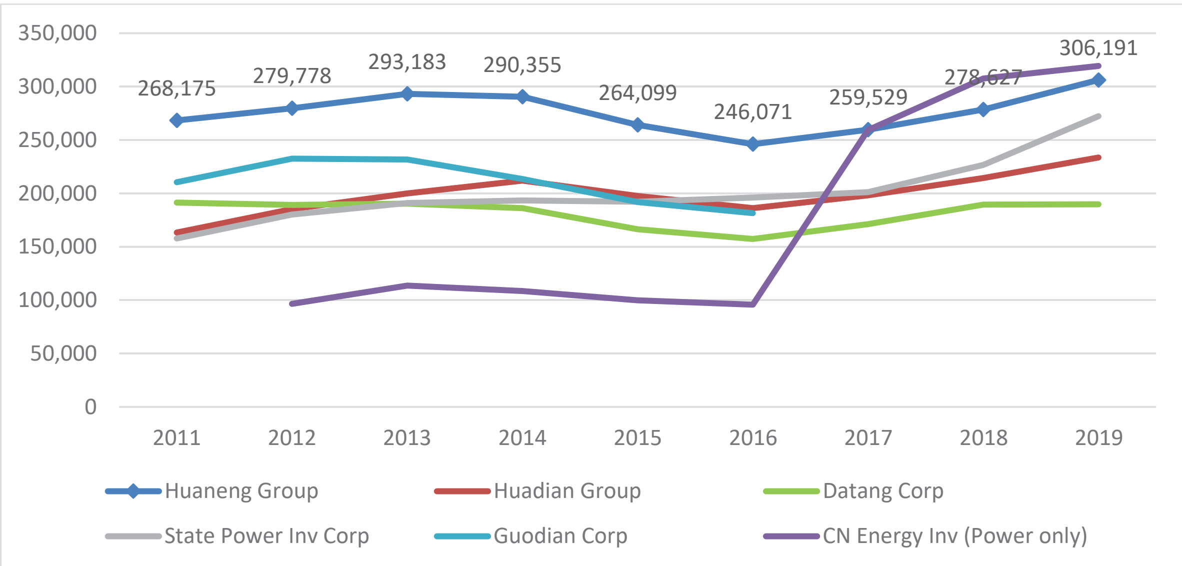
EXHIBIT 9: ASSETS AND REVENUE COMPARISON OF NON-LISTED PARENTS

Asset Comparison of Non-Listed Parents (in RMB Million)



Source: Constructed based on financial data obtained from Bloomberg Terminal, accessed February 2024.

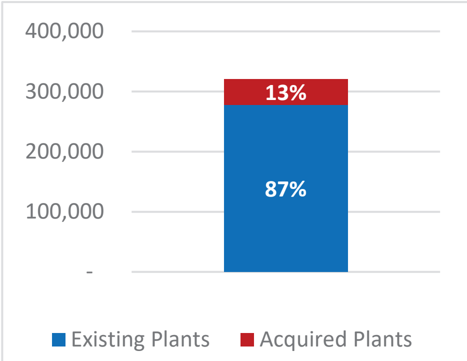
Revenue Comparison of Non-Listed Parents (in RMB Million)



Note: China Energy Investment Corp uses data from the Shenhua Group prior to 2017. Shenhua Group acquired Guodian Corp to form China Energy Investment Corp in 2017 (hence, there is no data for Guodian after 2016).

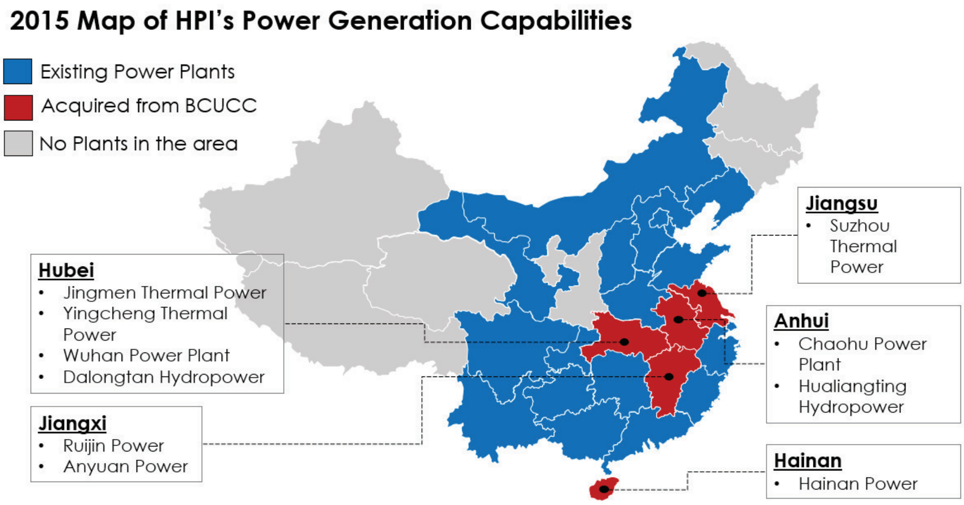
Source: Constructed based on financial data obtained from Bloomberg Terminal, accessed February 2024.

EXHIBIT 10: HPI: POWER GENERATION (IN GWH) BY ACQUIRED PLANTS VERSUS EXISTING PLANTS IN 2015



Source: Constructed based on information disclosed in the HPI's H-shares 2015 annual report, https://www.hpi.com.cn/sites/english/_layouts/15/WopiFrame.aspx?sourcedoc={61edff0a-9463-4e21-88dc-8b79bbb43c35}&action=interactivepreview, accessed February 2024.

EXHIBIT 11: MAP OF HPI'S EXISTING AND ACQUIRED POWER PLANTS IN MAINLAND CHINA IN 2015[38]



Source: Constructed based on information disclosed in the HPI's H-shares 2015 annual report and HPI's ADR 2015 Form 20-F filed with SEC, https://www.hpi.com.cn/sites/english/_layouts/15/WopiFrame.aspx?sourcedoc={61edff0a-9463-4e21-88dc-8b79bbb43c35}&action=interactivepreview, accessed February 2024.

38 The map shows the geographical concentration of HPI after the BCUCC and was constructed based on information in the HPI H share 2015 annual report.

EXHIBIT 12: HPI'S ACCOUNTING POLICIES ON BCUCC FOR REPORTING OF ITS DIFFERENT TYPES OF SHARES

HPI's A-shares: Traded on the Shanghai Stock Exchange (SSE), Ticker 6000011.	For the A-shares' financial reports (the reports were in Chinese), HPI adopted the Chinese Accounting Standards for Business Enterprises (CASBE, referred to as PRC GAAP in this case). Under PRC GAAP, HPI was required to apply the predecessor method to account for its BCUCC transaction.
HPI's H-shares: Traded at the Stock Exchange of Hong Kong (HKEX), Ticker 902.	For H-shares' financial reports (the reports were in English), HPI adopted IIFRS. Under IFRS, the receiving entity could choose the appropriate accounting method because there was no applicable IFRS standard for BCUCC. HPI chose to apply the acquisition method for all business combinations including the BCUCC.
HPI's ADR: Traded at New York Stock Exchange, Ticker HNP.	HPI adopted IFRS in its disclosure reports Form 20-F filed with the Securities Exchange Commission in the United States. HPI chose to apply the acquisition method for all business combinations including the BCUCC.

Source: HPI's H-shares 2015 annual report, HPI's A-shares 2015 annual report, and HPI's ADR 2015 Form 20-F filed with SEC, accessed February 2024

EXHIBIT 13: ITEMS REPORTED USING THE ACQUISITION METHOD UNDER IFRS REPORTING FOR THE BCUCC

Items reported under IFRS reporting	**Amounts expressed in thousands of RMB**
Fair value of total identifiable net assets	10,412,147
Non-controlling interests	(1,934,865)
Goodwill	1,169,966
Cash Consideration	9,647,248

Source: HPI's H-shares 2015 annual report, accessed February 2024

EXHIBIT 14: ITEMS REPORTED USING THE PREDECESSOR METHOD UNDER PRC GAAP FOR THE BCUCC

Items reported under PRC GAAP reporting	Amounts expressed in thousands of RMB
Carrying amount or book value of total identifiable net assets*	7,041,622
Non-controlling interests*	(1,246,926)
Goodwill	Nil
Cash Consideration	9,647,248

*The details of the carrying amount of total identifiable net assets and non-controlling interests for each of the 10 entities were shown in Table 1 of Appendix B.

Source: HPI's A-shares 2015 annual report, accessed February 2024